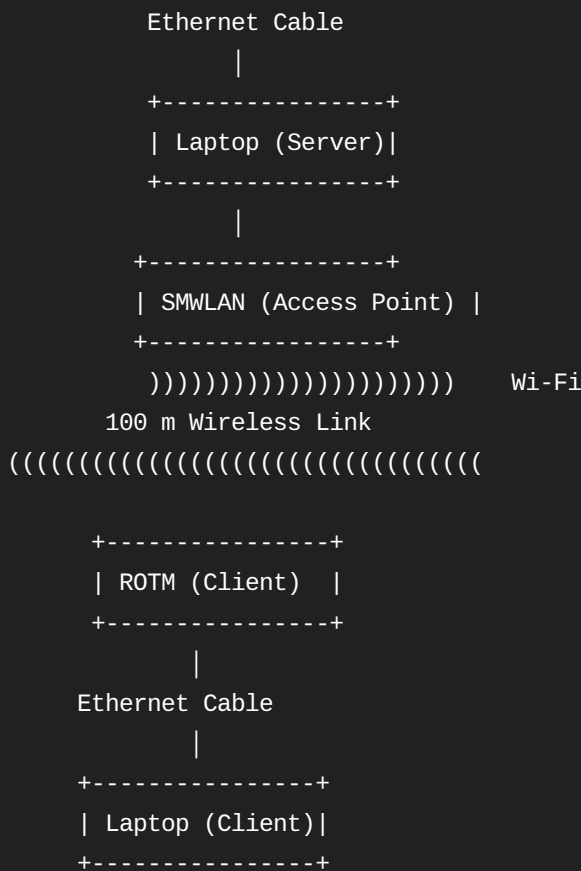


This is actually an excellent communication systems project because it combines **wireless communication, networking, IP protocols, MAC layer, routing, and throughput testing**. If you understand everything happening from powering on the devices until a successful ping or JPerf test, you'll be able to explain your project confidently in interviews.

Let's go through the entire process step by step.

Your Project Architecture



SMWLAN acts like a Wi-Fi router.

ROTM acts like a wireless station (client).

Step 1: Power ON

When you power both devices,

SMWLAN starts broadcasting.

It periodically sends **Beacon Frames**.

A beacon contains information like

- Network Name (SSID)
- Channel
- Supported data rates
- Security type
- Timing information
- MAC address of AP

Example

SSID : DefenceNet

Channel : 36

Security : WPA2

MAC : 00:11:22:AA:BB:CC

Every nearby client hears these beacon frames.

Step 2: Client Searches Networks

ROTM scans all channels.

There are two methods.

Passive Scan

It only listens.

Channel 1

↓

Beacon

↓

SSID Found

Active Scan

Client sends

Probe Request

AP replies

Probe Response

This is faster.

Step 3: Authentication

Now client says

"I want to join."

Authentication begins.

For WPA2

Client

↓

Authentication Request

↓

AP

↓

Authentication Response

Then comes the **4-way handshake**.

During this,

- Password is verified
 - Encryption keys are generated
 - Secure communication starts
-

Step 4: Association

Authentication only proves identity.

Association means

"You are now officially part of my network."

Association Request

↓

Association Response

Now AP stores

Client MAC Address

↓

Association ID

↓

Security Keys

↓

Capabilities

Now client becomes connected.

Step 5: DHCP or Static IP

Now laptop needs an IP.

Two possibilities.

Static IP

You manually configure

Laptop

192.168.10.2

Mask

255.255.255.0

AP

192.168.10.1

Done.

DHCP

If DHCP exists,

Client broadcasts

DHCP Discover

Server replies

DHCP Offer

Client

DHCP Request

Server

DHCP ACK

Now client has IP.

Step 6: MAC Address vs IP Address

This confuses many interview candidates.

Every device has

MAC

00:11:22:AA:BB:CC

and

IP

192.168.10.5

Think of it like

House number → IP

Person's fingerprint → MAC

MAC is used on the local network.

IP is used for logical addressing and routing.

Step 7: Ping

Suppose

Laptop A

```
192.168.10.2
```

Laptop B

```
192.168.10.3
```

You type

```
ping 192.168.10.3
```

Many things happen.

First ARP

Your laptop only knows

```
IP
```

```
192.168.10.3
```

But Wi-Fi sends frames using **MAC addresses**.

So it asks

```
Who has
```

```
192.168.10.3 ?
```

This is an **ARP Request**.

It is broadcast.

Every device receives it.

Only Laptop B replies

I am

192.168.10.3

My MAC

88:AA:44:11:77

Laptop A stores this in the ARP cache.

Step 8: ICMP

Now ping begins.

Ping uses **ICMP**.

Laptop A sends

Echo Request

Destination IP

192.168.10.3

Destination MAC

88:AA:44:11:77

Packet goes

Laptop

↓

Ethernet

↓

SMWLAN

↓

Wireless

↓

ROTM

↓

Ethernet

↓

Laptop

Laptop B replies

Echo Reply

That's why ping shows

Reply from

192.168.10.3

Time = 3 ms

Complete Packet Encapsulation

Before transmission, the data is wrapped with headers at each layer:

Application Data

|

▼

ICMP Header

|

▼

IP Header

|

▼

MAC Header

|

At the receiver, these headers are removed in reverse order to recover the original data.

Why Ping Time Changes

Propagation delay

-

Processing delay

-

Transmission delay

-

Queue delay

Total

RTT

Round Trip Time

What is JPerf?

JPerf is a graphical interface for **Iperf**, a network performance measurement tool.

It measures

- Throughput
- Bandwidth
- Packet loss (UDP)
- Jitter (UDP)
- Retransmissions (TCP)
- Connection quality

Ping only tells

Can I reach it?

JPerf tells

How good is the connection?

JPerf Working

One PC becomes

Server

Another becomes

Client

Server waits.

Client sends traffic.

Client

=====>

Server

Server measures received data.

TCP Test

Suppose

100 MB transmitted.

Time

10 seconds

Bandwidth

100 MB

÷

10

=

10 MB/s

JPerf displays this as throughput.

UDP Test

UDP has no acknowledgements.

JPerf reports

- Packet loss
- Jitter
- Delay

These metrics are important for voice, video, and real-time communication.

Why Defence Uses JPerf

Your project needs to verify that the wireless link is reliable.

For example, if the required throughput is **50 Mbps** but the measured throughput is **20 Mbps** at 100 meters, you know the link quality is not meeting expectations.

OSI Layers in Your Project

Layer	What Happens
Application	JPerf, Ping command
Transport	TCP (JPerf TCP) or UDP (JPerf UDP)
Network	IP addressing, ICMP packets
Data Link	MAC addressing, Wi-Fi frames, ARP
Physical	RF transmission between SMWLAN and ROTM

Questions You Should Be Ready to Answer

For your project, interviewers may ask:

- Why do we need an Access Point?
- What is the difference between AP mode and client mode?
- What is an SSID?
- What is the purpose of beacon frames?
- How does WPA2 authentication work?
- What is the difference between MAC and IP addresses?
- Why is ARP required before ping?
- What is ICMP and why does ping use it?
- How does a Wi-Fi frame differ from an Ethernet frame?
- What causes latency in a wireless link?
- Why does throughput measured by JPerf differ from the advertised data rate?
- What is jitter, and why is it important?
- Why use TCP versus UDP in JPerf?
- How do RSSI, SNR, and channel interference affect performance?
- How would you troubleshoot if the client cannot connect to the AP?

A recommended learning path for your project

- 1. Wi-Fi basics (IEEE 802.11):** SSID, channels, beacon frames, authentication, association.
- 2. Networking fundamentals:** MAC addresses, ARP, IP addressing, subnet masks, DHCP.
- 3. Protocols:** ICMP (ping), TCP, UDP, DNS (if used), routing basics.
- 4. Wireless performance:** RSSI, SNR, modulation, data rates, throughput vs. bandwidth.
- 5. Testing tools:** JPerf/Iperf, Wireshark (to observe ARP, ICMP, TCP/UDP packets), and the `ping` and `traceroute` commands.

Learning these topics in order will help you understand every stage of communication in your defence wireless project—from the moment the client discovers the access point to the point where JPerf measures the quality of the wireless link.

TCP (Transmission Control Protocol)

TCP is a **connection-oriented**, **reliable** transport layer protocol (Layer 4 of the OSI model). It ensures that data reaches the destination **accurately, in order, and without loss**.

How TCP Works

1. Connection Establishment (Three-Way Handshake)

Before sending data, TCP establishes a connection.

```
Client                                Server
| ----- SYN -----> |
| <--- SYN + ACK --- |
| ----- ACK -----> |
```

After this handshake, both devices are ready to communicate.

2. Data Transmission

TCP divides data into **segments**.

Each segment contains:

- Source Port
- Destination Port
- Sequence Number
- Acknowledgment Number
- Data
- Checksum

Example:

```
Segment 1 → Seq = 100
Segment 2 → Seq = 200
Segment 3 → Seq = 300
```

The receiver sends an ACK after receiving data.

3. Error Detection & Retransmission

If a segment is lost:

```
Sender → Segment 1 ✓  
Sender → Segment 2 ✗ (Lost)  
Sender → Segment 3 ✓  
  
Receiver:  
ACK for Segment 1  
No ACK for Segment 2  
  
Sender retransmits Segment 2
```

This makes TCP reliable.

4. Connection Termination

TCP closes the connection using a **four-way handshake** (FIN/ACK exchange).

TCP Features

- Connection-oriented
 - Reliable delivery
 - Error detection
 - Retransmission of lost packets
 - Flow control (prevents receiver overload)
 - Congestion control (prevents network overload)
 - In-order delivery
-

Advantages

- Reliable communication
 - No packet loss (unless the connection fails)
 - Data arrives in order
 - Suitable for large file transfers
-

Disadvantages

- More overhead

- Higher latency due to acknowledgments and retransmissions
 - Slightly slower than UDP
-

Applications of TCP

- Web browsing (HTTP/HTTPS)
 - Email (SMTP, IMAP, POP3)
 - FTP/SFTP
 - SSH
 - Database communication
-

UDP (User Datagram Protocol)

UDP is a **connectionless** transport layer protocol. It sends data without establishing a connection and does **not guarantee delivery or ordering**.

How UDP Works

There is **no handshake**.



```
graph LR; Client[Client] -- "Data Packet" --> Server[Server]
```

Client -----> Server
Data Packet

The sender transmits packets, but does not wait for acknowledgments.

UDP Packet (Datagram)

Contains:

- Source Port
- Destination Port
- Length
- Checksum
- Data

Compared to TCP, the header is much smaller.

Characteristics

- No connection setup
 - No acknowledgments
 - No retransmission
 - No sequencing
 - Lower overhead
 - Faster transmission
-

Advantages

- Very fast
 - Low latency
 - Small protocol overhead
 - Suitable for real-time communication
-

Disadvantages

- No guarantee of delivery
 - Packets may be lost
 - Packets may arrive out of order
 - No congestion control
-

Applications of UDP

- Video streaming
 - Voice over IP (VoIP)
 - Online gaming
 - DNS queries
 - Live broadcasting
-

TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable (best effort)
Handshake	Yes (3-way handshake)	No

Feature	TCP	UDP
Acknowledgment	Yes	No
Packet Order	Guaranteed	Not guaranteed
Retransmission	Yes	No
Speed	Slower	Faster
Header Size	20 bytes (minimum)	8 bytes
Overhead	Higher	Lower
Best For	File transfer, web, email	Voice, video, gaming, streaming

TCP and UDP in Your Defence Project

When using **JPerf (Iperf)**:

TCP Mode

- Measures **maximum reliable throughput**
- Retransmits lost packets
- Suitable for testing file-transfer-like performance

Example output:

```
Bandwidth: 92 Mbps
Retransmissions: 5
```

UDP Mode

- Measures:
 - Throughput
 - Jitter
 - Packet loss

Example output:

```
Bandwidth: 95 Mbps
Jitter: 1.2 ms
```

UDP mode is often used to evaluate **real-time communication performance**, while TCP mode evaluates **reliable data transfer performance**.

Interview Questions

1. Why is TCP reliable?

Answer: TCP uses sequence numbers, acknowledgments (ACKs), checksums, retransmissions, and flow/congestion control to ensure data is delivered correctly and in order.

2. Why is UDP faster than TCP?

Answer: UDP does not establish a connection, send acknowledgments, retransmit lost packets, or maintain packet ordering, so it has less overhead and lower latency.

3. Why does JPerf support both TCP and UDP?

Answer:

- **TCP** measures reliable throughput and retransmissions.
- **UDP** measures real-time network performance, including throughput, jitter, and packet loss.

4. Which protocol is used by the `ping` command?

Answer: Neither TCP nor UDP. `ping` uses **ICMP (Internet Control Message Protocol)**, which operates at the network layer.

5. Which protocol would you choose for your defence wireless link?

- **TCP:** When reliable delivery is required (e.g., command messages, configuration files).
- **UDP:** When low latency is more important than perfect reliability (e.g., voice, video, telemetry, or sensor streams where occasional packet loss is acceptable).

any things related to this please put

Yes. For a **Defence Wireless Communication Engineer** or **Embedded Communication Engineer**, there are many networking and wireless topics beyond TCP/UDP that you should know. Based on your SMWLAN-ROTM project, here is a complete learning checklist.

1. OSI Model (Most Important)

Layer	Protocols/Technologies	Your Project
Application	JPerf, Ping	Throughput testing
Transport	TCP, UDP	Data transfer
Network	IP, ICMP, ARP	Addressing and ping
Data Link	MAC, Wi-Fi (802.11)	Frame transmission
Physical	RF, Antenna, Modulation	Wireless communication

Interview Question: Why does the OSI model exist?

Answer: It divides communication into layers so each layer has a specific responsibility, making systems easier to design, troubleshoot, and maintain.

2. IEEE 802.11 (Wi-Fi Standard)

Know:

- 802.11a
- 802.11b
- 802.11g
- 802.11n
- 802.11ac
- 802.11ax (Wi-Fi 6)

Understand:

- Frequency bands (2.4 GHz, 5 GHz, 6 GHz)
- Channel width (20/40/80/160 MHz)

- Data rates
 - MIMO
 - OFDM
-

3. Access Point (AP)

An AP:

- Creates the wireless network
- Broadcasts the SSID
- Authenticates clients
- Encrypts traffic
- Bridges wireless and wired networks

Think of it as a traffic controller for all wireless clients.

4. Client (Station)

A client:

- Searches for APs
- Authenticates
- Associates
- Sends and receives data

In your project:

- **SMWLAN = Access Point**
 - **ROTM = Client**
-

5. SSID

SSID = Service Set Identifier

Example:

DEFENCE_NET

This is simply the network name broadcast by the AP.

6. BSSID

Every AP has a unique MAC address.

Example:

```
SSID : DEFENCE_NET
```

```
BSSID : 00:11:22:33:44:55
```

Many APs can broadcast the same SSID, but each has a different BSSID.

7. Beacon Frame

Every AP periodically sends beacon frames.

Beacon contains:

- SSID
- Channel
- Security type
- Supported rates
- Timestamp

Clients listen to these to discover available networks.

8. Authentication

Security process before joining.

Common methods:

- Open
- WPA2
- WPA3

During WPA2:

- Password verification
 - Encryption keys generated
-

9. Association

Authentication verifies identity.

Association makes the client an official member of the wireless network.

10. MAC Address

Example:

08:00:27:AB:CD:EF

- 48-bit hardware address
 - Layer 2
 - Used within the local network
-

11. IP Address

Example:

192.168.10.25

- Layer 3
 - Logical address
 - Used for routing between networks
-

12. ARP (Address Resolution Protocol)

Suppose you ping:

192.168.10.25

The computer needs the MAC address.

ARP asks:

Who has 192.168.10.25?

The destination replies:

I have it.
My MAC is 08:00:27:AB:CD:EF

Without ARP, communication on the local network cannot begin.

13. ICMP

Used for:

- Ping
- Error reporting
- Network diagnostics

Commands:

- `ping`
- `tracert`

14. DNS

Converts domain names into IP addresses.

Example:

`google.com`

↓

`142.250.x.x`

Your project may not use DNS if you work only with static IPs, but it is an important networking concept.

15. DHCP

Automatically assigns:

- IP address
- Subnet mask
- Gateway
- DNS server

In defence systems, static IPs are often preferred for predictability.

16. Subnet Mask

Example:

IP

192.168.1.10

Mask

255.255.255.0

It determines which devices are in the same network.

17. Default Gateway

Needed to communicate with devices outside the local network.

If both devices are on the same subnet, traffic does not need a gateway.

18. Routing

Routers decide where packets should go based on destination IP addresses.

19. Switching

A switch forwards Ethernet frames based on MAC addresses.

20. Throughput vs Bandwidth

Many people confuse these.

Bandwidth:

- Maximum theoretical capacity

Example:

100 Mbps

Throughput:

- Actual measured data transfer

Example:

JPerf measures throughput.

21. Latency

Time for data to travel from source to destination.

Measured in milliseconds.

22. Jitter

Variation in packet delay.

Example:

Packet 1 = 10 ms

Packet 2 = 20 ms

Packet 3 = 8 ms

High jitter affects:

- Voice
 - Video
 - Real-time control
-

23. Packet Loss

Packets that never reach the destination.

Causes:

- Weak signal
 - Interference
 - Congestion
-

24. RSSI (Received Signal Strength Indicator)

Measures received signal strength.

Example:

-40 dBm

Excellent

-65 dBm

Good

-80 dBm

Weak

25. SNR (Signal-to-Noise Ratio)

Measures how much stronger the signal is compared to noise.

Higher SNR means better communication.

26. Modulation

Examples:

- BPSK
- QPSK
- 16-QAM
- 64-QAM
- 256-QAM

Higher-order modulation:

- Higher speed
 - Requires better signal quality
-

27. AGC (Automatic Gain Control)

Automatically adjusts receiver gain.

If the signal becomes weak:

- Gain increases

If the signal becomes strong:

- Gain decreases
-

28. AMC (Adaptive Modulation and Coding)

Changes modulation and coding based on channel quality.

Poor signal:

QPSK

Good signal:

64-QAM

29. MIMO (Multiple Input Multiple Output)

Uses multiple antennas.

Benefits:

- Higher throughput
 - Better range
 - Improved reliability
-

30. OFDM (Orthogonal Frequency Division Multiplexing)

Splits data across many orthogonal subcarriers.

Advantages:

- High speed
- Resistant to multipath fading

Used in:

- Wi-Fi
- LTE
- 5G

31. Channel

Wi-Fi operates on channels.

Example:

- Channel 1
- Channel 6
- Channel 11

Choosing the correct channel reduces interference.

32. Interference

Sources:

- Nearby Wi-Fi networks
- Bluetooth
- Microwave ovens
- Other RF transmitters

Interference reduces throughput and increases packet loss.

33. Ping

Uses ICMP.

Checks:

- Reachability
- Round-trip time (RTT)

It does **not** measure maximum bandwidth.

34. Traceroute

Shows every router (hop) between source and destination.

Useful for identifying where delays occur.

35. JPerf / Iperf

Measures:

- TCP throughput
- UDP throughput
- Jitter
- Packet loss
- Retransmissions

Common commands:

Server:

Bash

```
iperf3 -s
```

Client:

Bash

```
iperf3 -c 192.168.10.2
```

UDP test:

Bash

```
iperf3 -u -c 192.168.10.2
```

36. Security

Know:

- WPA2
- WPA3
- AES encryption
- Passphrase authentication
- Four-way handshake

37. Troubleshooting (Very Important in Interviews)

If a client cannot connect:

1. Check power.

2. Verify AP is broadcasting the correct SSID.
 3. Check authentication credentials.
 4. Verify IP configuration (static/DHCP).
 5. Confirm subnet mask and gateway.
 6. Use `ping` to test connectivity.
 7. Check ARP table.
 8. Measure RSSI/SNR.
 9. Run JPerf to verify throughput.
 10. Check for RF interference or excessive distance.
-

38. Defence-Specific Concepts

Wireless links in defence systems are often evaluated for:

- Link reliability
 - Low latency
 - Secure communication
 - Long-range connectivity
 - Throughput under varying conditions
 - Packet loss and jitter
 - Operation in noisy RF environments
 - Encryption and authentication
 - Redundancy and failover
-

A Complete Learning Roadmap

Study these topics in this order:

1. OSI Model
2. TCP/IP Model
3. Ethernet and MAC
4. IP Addressing and Subnetting
5. ARP
6. ICMP (Ping)
7. TCP
8. UDP
9. DNS
10. DHCP
11. Routing and Switching
12. IEEE 802.11 (Wi-Fi)

- 13.** Authentication and Association
- 14.** RSSI, SNR, Noise, Interference
- 15.** Modulation (BPSK, QPSK, QAM)
- 16.** OFDM and MIMO
- 17.** AGC and AMC
- 18.** Throughput, Bandwidth, Latency, Jitter
- 19.** JPerf/Iperf
- 20.** Wireshark (packet analysis)
- 21.** RF fundamentals (path loss, antennas, fading)

Mastering these topics will allow you to explain **every stage of your SMWLAN-ROTM project**, from the moment the access point starts broadcasting until throughput is measured with JPerf, and will prepare you well for embedded communication and defence networking interviews.

ChatGPT can make mistakes. Check important info.